ALL MODELS ARE EQUAL, BUT SOME MODELS ARE MORE EQUAL THAN OTHERS

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ABSTRACT

As artificial intelligence systems evolve and integrate into our society, it becomes increasingly important to have well-designed evaluations to provide evidence about system capabilities and the risks that follow. However, a static benchmark inevitably loses its purpose over time as strong models receive similarly high scores on the benchmark. Creating high-quality tasks that can continue to distinguish model capabilities require substantial expert work, motivating other ways to continue extracting capability signals besides the final evaluation score. We thus introduce a new problem setting of *scalable capability evaluations*, where we study whether capability beyond the current benchmark frontier can be predicted from saturated tasks. We retrospectively evaluate representative approaches for extracting this signal such as pairwise LLM judgments of model transcripts, single-transcript absolute scoring, transcript- and metadata-based predictive models, and historical forecasting-style baselines. We find that [TODO: headline result; 1 2 lines?Lorem ipsum dolor sit amet, consectetur adipiscing elit.Donec imperdiet lacus in condimentum tincidunt.]. We also find that [TODO: qualitative result; 1 2 lines?Lorem ipsum dolor sit amet, consectetur adipiscing elit.] Our results suggest that saturated benchmarks can still retain useful capability information beyond accuracy scores, providing a potential way to extend the useful lifetime of existing evaluations.

1 INTRODUCTION

Capability evaluations estimate what an artificial intelligence system can do using a set of tasks and an evaluation protocol, allowing researchers to track AI progress and identify areas that may require stronger safeguards. Current frontier safety frameworks use these evaluations to classify risks and inform deployment decisions, including risks from biological and chemical applications, cybersecurity, autonomous research, and other high-risk domains (OpenAI, 2025; Anthropic, 2026; Flynn et al., 2026).

The value of a benchmark thus depends on its ability to distinguish between models. As model capabilities increase, score differences between state-of-the-art models shrink, until these differences become too small to support reliable conclusions. We consider a benchmark to be *saturated* when its scores no longer reliably discriminate between models of meaningfully different capabilities. Benchmark saturation has already been shown to affect current evaluations, with a recent study of 60 language model benchmarks finding signs of saturation in nearly half (Akhtar et al., 2026).

One way to address saturated benchmarks is to build harder evaluation tasks. Researchers designed benchmarks such as Humanity's Last Exam and MMLU-Pro after existing benchmarks became too easy for frontier models (Center for AI Safety et al., 2026; Wang et al., 2024b). However, the time and cost required for task construction and verification make continual benchmark updates difficult to match frontier model progress. This creates a recurring gap between when an existing benchmark becomes saturated and when a sufficiently difficult replacement becomes available.

We thus study whether useful capability information can still be extracted during this gap. Consider a benchmark at time t, when current frontier models are approaching saturation on this benchmark. When a new model is released, the new model's score lies outside of the benchmark's discriminative

range. We ask whether it is still possible to predict its performance on hypothetical harder tasks beyond the current benchmark ceiling.

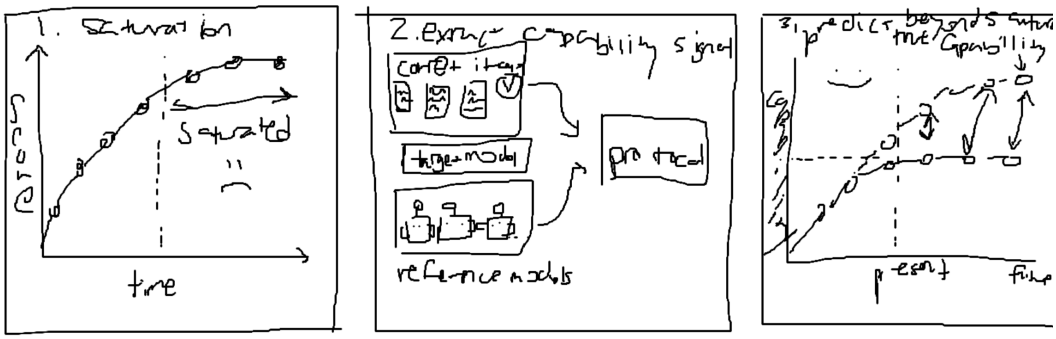


Figure 1: **Overview of our evaluation setting.** As benchmark performance saturates, we are no longer able to track model progress across time. We thus present a new problem setting of *scalable capability evaluations* that analyzes model behavior on saturated items together with earlier reference models to estimate the capability of a target frontier model. We then evaluate whether this estimate predicts performance on harder, unsaturated items.

Predicting capability beyond benchmark saturation is non-trivial for two main reasons. First, the predictable relationship between observable features and actual capability may change as model paradigms evolve. We have already seen one example of this with reasoning models which were able to substantially improve performance on difficult reasoning tasks using additional inference-time computation (OpenAI, 2024). Moreover, model transcripts specifically contain many potential confounders such as response length or model-family-specific patterns which may incorrectly correlate with capability. A useful prediction protocol must therefore extract signals that remain robust despite both distribution shift and these potentially confounding features.

We evaluate several representative approaches for extracting such signals. These include pairwise LLM judging of model transcripts, absolute transcript scoring, methods based on deterministic transcript or model metadata features, and historical forecasting-style baselines. To avoid allowing protocols to recover capability through ordinary correctness differences among weaker models, our primary evaluation restricts their inputs to responses for which the evaluated models reach the correct outcome.

We evaluate these methods retrospectively across six-month periods. At each time period, protocols are only given access to information that would have been available during that time period, such as previously released models. The prediction target is the performance of newly released frontier or near-frontier models on harder tasks beyond the saturated region. We construct this target using Item Response Theory (IRT) to separate tasks according to difficulty and test whether estimates derived from the saturated subset of a benchmark predicts performance for the next capability frontier.

Main findings.

1. **Capability beyond saturation can be predicted from solved-task behavior.** [TODO: Headline result. Report the primary capability-prediction metric across domains and compare against benchmark-score / forecasting baselines.]
2. **Prediction quality depends strongly on the information available to the evaluator.** [TODO: Headline result on pairwise judging, absolute judging, transcript features, metadata, CoT/no-CoT, or other major protocol comparison. State which signals account for most of the predictive performance.]
3. **Performance at the frontier and across time reveals the main limitation of current approaches.** [TODO: Headline result on frontier-only targets and temporal generalization. State whether estimates remain calibrated for models released after the references/judge, and highlight systematic underprediction if present.]

2 PROBLEM SETTING

We consider a setting in which model capabilities are improving at a faster rate than new benchmarks can be developed and validated. When benchmark scores no longer discriminate amongst the highest performing models, we consider the benchmark to be *saturated*.

For models that score below the point where the benchmark starts to saturate, ordinary benchmark scoring would remain a useful source of information about their capability. We focus on the complementary case in which a newly evaluated frontier or near-frontier model can no longer be reliably distinguished using benchmark scores alone.

2.1 SCALABLE CAPABILITY EVALUATIONS

We define *scalable capability evaluations* as a task of estimating a model's capability beyond the range that can be measured reliably by existing benchmarks. At a given *reference period* t, a set of reference models is available for which both behavior on the existing benchmark and estimates of their capability are known. At a later *target period* $t + 1$ a new *target model* is evaluated on the existing benchmark.

A *protocol* is any procedure that uses information available from the reference models, the target model, and the existing benchmarks to produce an estimate of the target model's capability.

3 EXPERIMENTAL METHODS

We instantiate the scalable capability evaluation setting using retrospective model releases and benchmark results. For each reference period t, we provide each protocol the information that would have been available at that time along with a set of models released in the subsequent target period $t + 1$, and compare its predictions against capability.

3.1 EVALUATION SETUP

Temporal splits. We divide model releases into consecutive six-month periods. For each prediction task, the *reference set* is made of models available up to period t, while the *target set* is made of models released during period $t + 1$.

Item calibration and saturation boundary. To study the post-saturation setting, we need a way to separate saturated benchmark items from harder items. However, defining this split from raw benchmark scores is difficult because each capability domain combines multiple benchmarks with different difficulties and formats. We therefore fit a global *Item Response Theory (IRT)* model within each domain to place all models and items on a common scale, which we then use to separate the saturated and unsaturated item sets.

We use a one-parameter logistic (1PL) IRT model, meaning that each model m is assigned a single capability parameter θ_m and each item a single difficulty parameter β_i and the probability that a model answers an item i correctly depends on the difference between these two values:

$$P(y_{mi} = 1) = \sigma(\theta_m - \beta_i),$$

where σ is the logistic function. For each target period, we use eight models from the reference period as reference models and eight models from the current period as target models. We define all items that are answered correctly by all three of the three highest-capability reference models as the *saturated set* and the rest as the *unsaturated set*. Protocol inputs are drawn only from the saturated items that the evaluated model itself answers correctly to completely isolate the protocols from correctness-derived signals.

Prediction target. To measure the capability beyond the saturated portion of the benchmark, we fit a second local 1PL IRT model to the reference and target models on the unsaturated set whil keeping ach item's difficulty parameter β_i fixed. We use the capability estimate θ^*_m estimated for each target model by the local IRT model as the prediction target.

Domains. We evaluate scalable capability estimation across four main domains: science, mathematics, coding, and cybersecurity. Each domain combines multiple benchmarks that span a range of item difficulties, allowing us to identify a saturated region and evaluate predictions on harder items beyond this region. [TODO: put actual benchmark names]

Model selection. We restrict our evaluation to open-weight models due to transcript-based protocols requiring access to the evaluated model's reasoning traces and final responses which were generally not exposed consistently by closed-source model APIs. [TODO: point to appendix for full list of models]

3.2 Prediction Protocols

We compare several ways of extracting capability information from the observations available under this setting.

Pairwise transcript comparison. For each reference period, the best LLM judge from the period receives responses from two models to the same item and predicts which response provides evidence of the model of higher capability. Judgments are aggregated across items and model pairs into Bradley-Terry model...

Single-transcript absolute scoring. An LLM judge evaluates one response at a time and assigns a capability-related score without observing another model's response. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris semper turpis ut libero blandit pharetra. Praesent mauris ante, posuere a accumsan eget, euismod ut enim. Donec vestibulum odio sit amet dignissim auctor. Fusce gravida elit et justo semper, venenatis aliquam dui interdum.

Transcript-feature models. We extract predefined deterministic features from each response, aggregate them across items, and fit a predictive model using the reference models. Nam in pretium magna, nec aliquet lectus. Pellentesque non lobortis tortor. Duis euismod orci et augue vehicula, nec iaculis erat rutrum. Nulla facilisi. Suspendisse potenti. Duis aliquam magna efficitur volutpat tincidunt. Nulla eu libero cursus, vestibulum turpis non, bibendum erat. Suspendisse sed sagittis enim. Mauris blandit odio ac ligula lacinia rutrum.

Model-metadata baseline. We predict capability using model-level information rather than benchmark responses. Mauris suscipit dui et maximus laoreet. Cras neque urna, sagittis at arcu a, tincidunt volutpat tortor. Phasellus tincidunt purus eu molestie efficitur. Sed eget odio elementum, interdum quam id, malesuada ligula.

Historical forecasting baseline. We predict target-model capability from trends among previously released models without using target-model transcripts. Morbi non massa ut nulla sagittis hendrerit nec vitae neque. In facilisis, tortor eu bibendum aliquam, nisi enim condimentum lectus, at imperdiet justo justo ut est. Nunc finibus feugiat mi, sit amet volutpat neque ultrices eget.

Conventional benchmark baseline. We use the target model's conventional benchmark score as a baseline for the capability information available from ordinary benchmark performance. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In pharetra augue dolor, eu bibendum ligula lobortis non. Curabitur consectetur at ligula at molestie.

[TODO: Specify high level details per protocol. point to appendix for more detail]

3.3 Calibration and Evaluation

[TODO: Specify the calibration function.]

Our primary metric is the mean squared error (MSE) between predicted and measured target model capability.

4 RESULTS

Can capability beyond benchmark saturation be predicted? We first evaluate protocols to see if they can recover capability information about the target models.

Table 1 reports mean squared error (MSE) for each protocol across capability domains and in aggregate. We compare transcript-based protocols against baselines that use only historical trends, model metadata, or conventional benchmark accuracy.

Table 1: **Prediction of capability beyond benchmark saturation.** Mean squared error between predicted capability and the target capability θ^* on the unsaturated set of benchmark items. Lower is better. Results are reported separately by domain and averaged across domains.

Protocol	Aggregate	Science	Math	Coding	Cyber
Pairwise transcript comparison	–	–	–	–	–
Absolute transcript scoring	–	–	–	–	–
Transcript features	–	–	–	–	–
Model metadata	–	–	–	–	–
Historical forecasting	–	–	–	–	–
Conventional benchmark accuracy	–	–	–	–	–

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Which information sources contain predictive signal? We next compare protocols according to the information available to them. Pairwise comparison and absolute scoring allow an LLM judge to inspect the semantic content of model responses, whereas transcript-feature models use only predefined surface-level features. Historical forecasting does not observe the target model's responses at all, while conventional benchmark accuracy uses only correctness information.

This comparison allows us to distinguish whether predictive performance comes primarily from the semantic content of solved responses, simple behavioral features, or trends that could have been inferred without inspecting model behavior.



Figure 2: **Predictive performance by information source.** Comparison of pairwise transcript judging, absolute transcript scoring, deterministic transcript features, historical forecasting, and conventional benchmark accuracy. Lower MSE indicates more accurate prediction of capability on hard items.

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Does prediction hold for frontier models? Our primary motivation is capability estimation for newly released frontier or near-frontier systems, rather than retrospective prediction of arbitrary models. We therefore separately evaluate prediction error on the highest-capability target models in each period.

Table 2: **Prediction performance on frontier and non-frontier target models.** We report MSE separately for frontier or near-frontier targets and the remaining target models.

Protocol	Frontier	Non-frontier
Pairwise transcript comparison	–	–
Absolute transcript scoring	–	–
Transcript features	–	–
Historical forecasting	–	–
Conventional benchmark accuracy	–	–

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Does prediction remain calibrated over time? A useful scalable evaluation protocol must remain informative as model capabilities and model families change. We therefore examine predictions separately across temporal splits rather than averaging over all periods.

Figure 3 compares predicted and measured capability for target models released in each period. Perfectly calibrated predictions would lie on the diagonal.



Figure 3: **Calibration across model-release periods.** Predicted capability is plotted against target capability θ^* for each temporal split. The dashed diagonal denotes perfect calibration. Points below the diagonal correspond to systematic underprediction of capability.

[TODO: Describe temporal generalization. In particular, report whether errors grow for later model generations or after major model-paradigm shifts, and whether any protocols systematically underpredict frontier capability. Mauris suscipit dui et maximus laoreet. Cras neque urna, sagittis at arcu a, tincidunt volutpat tortor. Phasellus tincidunt purus eu molestie efficitur. Sed eget odio elementum, interdum quam id, malesuada ligula. Morbi non massa ut nulla sagittis hendrerit nec vitae neque.

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4.1 Ablations

We perform several additional analyses to identify which choices in the evaluation setup drive prediction performance.

Correct-only versus all responses. Our primary setting exposes protocols only to responses for which the evaluated model reaches the correct outcome, preventing ordinary correctness differences from directly revealing model capability. We compare this against a less restrictive setting in which protocols receive all responses.

[TODO: Report whether access to incorrect responses substantially improves prediction. If so, quantify how much of the signal can be attributed to ordinary correctness differences rather than behavior within correct responses. Fusce eleifend tellus a leo dignissim lobortis. Morbi nisi lectus, interdum sed condimentum eget, efficitur in ex. Ut eu fermentum magna, ut sodales dolor. Mauris vulputate nisi fermentum, ornare sapien et, auctor nibh. Suspendisse nec pharetra urna, eget varius mi. Duis id erat ut lacus auctor fringilla sed eget libero. Ut hendrerit sapien et nunc venenatis, sed viverra leo ultricies.]

Table 3: **Sensitivity to evaluation setup.** Aggregate prediction error under alternative protocol inputs and prediction targets. Lower is better.

Setting	MSE	Δ
Correct only	–	–
All responses	–	–
Full hard set	–	–
Next difficulty band	–	–



Figure 4: **Scaling with evaluation budget.** Prediction error as the number of reference models or judged items increases. Curves show [TODO: protocols]; lower is better.

Prediction horizon beyond saturation. Our main target estimates capability using the full hard item set.

[TODO: Compare performance on the full hard set against the next difficulty band. If performance improves substantially for the local target, interpret this as evidence that prediction quality decreases with extrapolation distance. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris semper turpis ut libero blandit pharetra. Praesent mauris ante, posuere a accumsan eget, euismod ut enim. Donec vestibulum odio sit amet dignissim auctor. Fusce gravida elit et justo semper, venenatis aliquam dui interdum. Nam in pretium magna, nec aliquet lectus. Pellentesque non lobortis tortor. Duis euismod orci et augue vehicula, nec iaculis erat rutrum. Nulla facilisi. Suspendisse potenti. Duis aliquam magna efficitur volutpat tincidunt. Nulla eu libero cursus, vestibulum turpis non, bibendum erat. Suspendisse sed sagittis enim. Mauris blandit odio ac ligula lacinia rutrum.]

Frontier versus non-frontier targets. Finally, we test whether aggregate performance is driven primarily by easier prediction cases. We partition target models by capability and separately evaluate prediction error on frontier or near-frontier models and on lower-capability targets.

[TODO: Report the frontier gap. If prediction error increases toward the frontier, state whether this reflects systematic underprediction, increased variance, or both. Connect this result to Section 4. Ut sit amet justo efficitur, efficitur libero in, vulputate risus. Morbi aliquet velit arcu, nec ullamcorper lectus dignissim non. Mauris sed leo ut ligula sodales mollis sit amet vel ligula. Duis nec mi ac felis ultrices porttitor ut eget ex. Cras rutrum, quam quis mattis sodales, velit metus malesuada massa, sit amet tempor dolor dui non tellus. Phasellus lobortis dictum laoreet. Sed vehicula bibendum

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4.2 Additional Analysis

We next investigate what information in model responses drives successful capability prediction and whether this information generalizes across evaluation settings.

What information in the transcript is predictive? Transcript-based protocols may exploit several different signals, including the semantic content of the reasoning, properties of the final answer, or simple characteristics such as response length.

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Figure 5: **Error by target-model capability.** Prediction error for frontier or near-frontier targets compared with lower-capability target models. [TODO: Define frontier cutoff in caption or Methods.]

Is transcript length itself predictive? More capable models may produce systematically different response lengths, creating a simple signal that transcript-based protocols could exploit without understanding response content.

[TODO: Report the correlation between transcript length and capability, and the performance of a length-only baseline. Mauris suscipit dui et maximus laoreet. Cras neque urna, sagittis at arcu a, tincidunt volutpat tortor. Phasellus tincidunt purus eu molestie efficitur. Sed eget odio elementum, interdum quam id, malesuada ligula. Morbi non massa ut nulla sagittis hendrerit nec vitae neque. In facilisis, tortor eu bibendum aliquam, nisi enim condimentum lectus, at imperdiet justo justo ut est. Nunc finibus feugiat mi, sit amet volutpat neque ultrices eget. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In pharetra augue dolor, eu bibendum ligula lobortis non. Curabitur consectetur at ligula at molestie.]

5 Related Work

Benchmark saturation and evaluation design. Previous works on benchmark report the impact of score saturation and the downstream limit to discrimination amongst strong models (Bowman & Dahl, 2021; Vania et al., 2021; Akhtar et al., 2026). Many works address and attempt to resolve this by introducing new challenging versions of existing benchmarks (Hendrycks et al., 2021; Suzgun et al., 2023; Rein et al., 2024; Wang et al., 2024b; Center for AI Safety et al., 2026). Another line of works also explore the idea of dynamic evaluation which instead uses adversarial data collection or some continuously updated test set to maintain difficulty over time (Kiela et al., 2021; White et al., 2025; Jain et al., 2024), but costs for running these methods are much higher than alternatives.

Information from model responses. Previous works on transcript-analysis extracts signals beyond answer correctness through pairwise comparison and rubric-based scoring (Zheng et al., 2023; Liu et al., 2023; Kim et al., 2024), process-level reasoning assessment (Uesato et al., 2022; Lightman et al., 2024), and confidence-based prediction of correctness (Lin et al., 2022; Kadavath et al., 2022; Tian et al., 2023). These approaches suggest that model transcripts are a promising source of information. However, LLM judges have been reported to exhibit systematic response-length and

position biases (Zheng et al., 2023; Wang et al., 2024a; Dubois et al., 2024), suggesting signals can be sensitive to factors unrelated to actual capability.

Capability estimation and prediction. Previous works on Item Response Theory (IRT) uses it for latent ability estimation, item calibration, and test-set comparison (Lalor et al., 2016; 2019; Rodriguez et al., 2021; Vania et al., 2021; Zhou et al., 2026). By accounting for differences in item difficulty, IRT can be used to provide a more structured capability estimate than accuracy. Additional work on adaptive testing and representative item selection focus on a different direction of reducing evaluation cost by identifying informative subsets of tasks (Zhuang et al., 2023; Maia Polo et al., 2024; Fonseca Rivera et al., 2026), although estimation reliability depends on the number and distribution of evaluated models (Jiang et al., 2026). Finally, forecasting work predicts benchmark performance from signals on other tasks (Owen, 2024; Ruan et al., 2024).

6 Discussion

We studied whether capability information beyond ordinary benchmark correctness can be recovered from model behavior on tasks that models already solve correctly. Our results suggest that [TODO: headline result], showing that benchmark saturation does not necessarily imply that all capability-relevant information has been exhausted.

More broadly, benchmark saturation is a measurement problem for the frontier. When conventional ways to evaluate lose their ability to distinguish between similarly high-scoring models, scalable capability evaluation may help maintain visibility into capability growth and provide additional evidence for governance decisions. In this way, we believe scalable capability evaluation to be complementary to constructing harder benchmarks, as it may help preserve visibility during the period after existing evaluations lose resolution but before suitable replacements are available.

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